



BIOLOGY HSSC-I **SECTION – A (Marks 17)**

Time allowed: 25 Minutes

Section – A is compulsory. All parts of this section are to be answered on this page and handed over to the Centre Superintendent. Deleting/overwriting is not allowed.

Do not use lead pencil.

حصہ اول لازمی ہے۔ اس کے جوابات اسی صفحہ پر دے کر نام مرکز کے حوالے کریں۔ کاٹ کر دوبارہ لکھنے کی اجازت نہیں ہے۔ لے پیل کا استعمال ممنوع ہے۔

Version No.				
3	0	0	6	1

ROLL NUMBER						

0	●	●	0	0
1	1	1	1	●
2	2	2	2	2
●	3	3	3	3
4	4	4	4	4
5	5	5	5	5
6	6	6	●	6
7	7	7	7	7
8	8	8	8	8
9	9	9	9	9

0	0	0	0	0	0	0
1	1	1	1	1	1	1
2	2	2	2	2	2	2
3	3	3	3	3	3	3
4	4	4	4	4	4	4
5	5	5	5	5	5	5
6	6	6	6	6	6	6
7	7	7	7	7	7	7
8	8	8	8	8	8	8
9	9	9	9	9	9	9

Answer Sheet No. _____

ہر سوال کے سامنے دیے گئے، کرکیم لیم کے مطابق درست دائرہ کو پر کریں۔

Invigilator Sign. _____

Fill the relevant bubble against each question according to curriculum:

Candidate Sign. _____

Question	A	B	C	D	A	B	C	D
1. The control center of the cell is:	Mitochondria	Nucleus	Ribosome	Chloroplast	☐	☐	☐	☐
2. Which of the following is NOT a carbohydrate?	Glucose	Starch	Hemoglobin	Cellulose	☐	☐	☐	☐
3. Which polysaccharide serves as the main structural component in plant cell walls?	Starch	Cellulose	Glycogen	Chitin	☐	☐	☐	☐
4. Which model explains enzyme action by showing a change in the active site to fit the substrate?	Lock and Key Model	Induced Fit Model	Michaelis-Menten Model	Transition State Model	☐	☐	☐	☐
5. Which of the following best describes the role of ATP synthase in cellular respiration?	Splits glucose into pyruvate	Transports electrons along ETC	Utilizes proton gradient to form ATP	Transfers acetyl group to oxaloacetate	☐	☐	☐	☐
6. HIV primarily attacks which type of human cells?	Red blood cells	B lymphocytes	T-helper cells	Platelets	☐	☐	☐	☐
7. Which structure is present in bacteria for movement?	Pili	Capsule	Flagellum	Cilia	☐	☐	☐	☐
8. Mycorrhiza represents which type of ecological relationship?	Parasitism	Commensalism	Competition	Mutualism	☐	☐	☐	☐
9. Which group of plants is characterized by the absence of vascular tissue?	Pteridophytes	Gymnosperms	Angiosperms	Bryophytes	☐	☐	☐	☐
10. Which plant tissue is responsible for the transport of sugars?	Phloem	Xylem	Collenchyma	Sclerenchyma	☐	☐	☐	☐
11. Which phylum is characterized by a water vascular system used in locomotion and feeding?	Annelida	Arthropoda	Mollusca	Echinodermata	☐	☐	☐	☐
12. Which hormone regulates the development of male secondary sexual characteristics?	Testosterone	Progesterone	Estrogen	FSH	☐	☐	☐	☐
13. Which blood group results from the presence of both I^A and I^B alleles?	A	B	O	AB	☐	☐	☐	☐
14. Which genetic cross illustrates incomplete dominance?	Red flower × White flower → Pink flower	Red flower × White flower → All red	Tall pea plant × Dwarf pea plant → All tall	Yellow seed × Green seed → All yellow	☐	☐	☐	☐
15. The semi-conservative model of DNA replication was experimentally proven by:	Watson and Crick	Hershey and Chase	Meselson and Stahl	Griffith	☐	☐	☐	☐
16. The presence of wings in bats and insects, used for flight but with different structural origins, is an example of:	Homologous structures	Analogous structures	Vestigial structures	Adaptive radiation	☐	☐	☐	☐
17. Animal in which blastopore develop into mouth are called:	Pseudocoelomates	Acoelomates	Protostomes	Chordates	☐	☐	☐	☐

**BIOLOGY HSSC-I****Time allowed: 2:35 Hours****Total Marks Sections B and C: 68****SECTION – B (Marks 42)****Q. 2 Answers the following parts briefly.****(14 x 3 = 42)**

(i)	How mitochondria developed according to endosymbiont theory? Briefly explain.	03	OR	Draw ultrastructure of an animal cell and label three organelles which are present only in animal cell.	03												
(ii)	Compare amylase starch with amylopectin on the basis of structures, linkage and solubility.	1x3	OR	Briefly describe the composition and role of 'capsule' in bacterial cell.	2+1												
(iii)	Distinguish between saturated and unsaturated fatty acids with examples.	1.5 + 1.5	OR	What are three types of cofactors? Give one example of each.	1x3												
(iv)	Identify the role of active site of enzyme. Also mention its component parts.	1.5 + 1.5	OR	Briefly explain hormonal control of Sperm production in negative feedback system.	03												
(v)	Briefly explain importance of competitive inhibitors.	03	OR	Complete the table: <table><tr><th>Specimen</th><th>Name of sub-division</th><th>One special characteristic</th></tr><tr><td>Rhynia</td><td></td><td></td></tr><tr><td>Lycopodium</td><td></td><td></td></tr><tr><td>Equisetum</td><td></td><td></td></tr></table>	Specimen	Name of sub-division	One special characteristic	Rhynia			Lycopodium			Equisetum			03
Specimen	Name of sub-division	One special characteristic															
Rhynia																	
Lycopodium																	
Equisetum																	
(vi)	Briefly explain three steps of oxidation of pyruvate into acetyl Co-A.	1x3	OR	Why are the chances of a husband and wife having either a boy or a girl 50:50?	03												
(vii)	Complete the table: <table><tr><th></th><th>Site 1</th><th>Site 2</th><th>Site 3</th></tr><tr><td>Proton pumping site</td><td>FMn Co Q</td><td>Cyto.b...</td><td>Cyto – a</td></tr><tr><td>Enzyme involved</td><td></td><td>Cyto..C₁</td><td>Cyto – a₃</td></tr></table>		Site 1	Site 2	Site 3	Proton pumping site	FMn Co Q	Cyto.b...	Cyto – a	Enzyme involved		Cyto..C ₁	Cyto – a ₃	03	OR	Draw labelled life cycle of HIV.	03
	Site 1	Site 2	Site 3														
Proton pumping site	FMn Co Q	Cyto.b...	Cyto – a														
Enzyme involved		Cyto..C ₁	Cyto – a ₃														
(viii)	Name three types of enzymes for HIV replication and write role of each.	1x3	OR	What is heat of vaporization? What is its importance for animals?	1+2												
(ix)	Briefly explain any three prokaryotic features of cyanobacteria.	1x3	OR	Differentiate between incomplete dominance and codominance.	1.5 + 1.5												
(x)	Illustrate with diagrams three types of rod shape bacteria.	1x3	OR	Draw a labeled Calvin cycle.	2+1												
(xi)	Enlist any six stages of endospore formation in bacteria as a mechanism of survival to withstand unfavourable conditions.	03	OR	Classify animals on the basis of Body cavity with one example of each.	1x3												
(xii)	Describe genomic organization of bacteria with respect to circular DNA and plasmids.	1.5 + 1.5	OR	Why post transcriptional modification of mRNA is necessary for eukaryotes? Give at least three reasons.	1x3												
(xiii)	Write down names and causative agents of any three fungal diseases in man.	1x3	OR	Briefly describe population, community and ecosystem.	1x3												
(xiv)	Describe any three general features of vascular plants.	1x3	OR	Briefly state three sources of mutation with examples.	03												

SECTION – C (Marks 26)**Attempt the following questions.**

Q.3	Explain formation and functions of lysosomes with suitable sketch	2+3 +2	OR	Draw pay off phase of Glycolysis pathway and explain its steps.	2+5
Q.4	Explain double helical structure of DNA with suitable sketch as proposed by Watson and Crick.	5+2	OR	Explain steps of infection process of bacteriophage and replication of bacteriophage in lysogenic cycle.	3+4
Q.5	Explain components of TACT force and its on mechanism.	4+2	OR	Explain four steps of Griffith experiment and conclusions to prove DNA hereditary material.	4+2
Q.6	Explain inheritance of wheat grain colour. Make a suitable checker board as an example of Polygenic inheritance.	3+3	OR	What is greenhouse effect? Explain any five impacts of greenhouse gases on climate change.	1+5